When the Public Leaderboard Misleads: Lessons from 95 Scored Submissions in Single-Subject MEG Word Decoding

Tun-Yu Kuo
Independent Researcher
kuotunyh@gmail.com
Kaggle team: yu_kuo14

Abstract

We describe our entry to the Deep track of the 2026 PNPL Competition, in which systems use magnetoencephalography (MEG) recordings to predict which of 50 English words a participant hears; submissions are scored by class-balanced top-10 accuracy (BAcc@10). Our system combines an ensemble of convolutional word encoders and sentence-level Transformers with a closed-vocabulary language-model (LM) prior. Kaggle's private leaderboard ranked each team by the best private score among up to ten submissions it selected. In the last two days, an in-domain LM fine-tuned on synthetic vocabulary-only sentences and whole-sentence candidate decoding raised our public score to 0.733 (13th of 90 teams). On the private split, however, the best of our ten selected submissions scored 0.663 (44th), while an earlier, simpler submission scored 0.688. Using the public and private scores of our 95 pre-deadline submissions, we examine this gap. Among those with public score at least 0.66, the 24 made after the in-domain LM was introduced averaged 0.714 public and 0.648 private, against 0.686 and 0.666 for the 36 earlier ones. Re-seeding only the fine-tuning of the in-domain LM changed the private score by up to 0.015, more than most effects we tested, and only 10 of 26 later changes moved the two scores in the same direction. These are exploratory observations from one development history, from which we draw practical recommendations for choosing final submissions when the public split is small.

1 Introduction

In the Deep track, all data come from one participant (`subj0`). The training data are MEG recordings of about 780k spoken words from audiobooks, TIMIT, MOCHA-TIMIT and podcasts. Each test item is a word the participant heard, either within a sentence or in isolation. For a test sentence the organisers provide its MEG segment and the onset of every word, so the number and order of words are known; isolated words come as 1 s epochs. The test sentences are short conversational utterances made almost entirely of the 50 target words [2]. For each test item a system outputs a probability distribution over the 50 words. A word's Recall@10 is the fraction of its occurrences for which it is among the ten most probable words, and BAcc@10 averages this over the 50 words, so chance is 0.20.

During the competition, Kaggle shows a public score computed on part of the test set. Each team then selects up to ten submissions, and Kaggle's private leaderboard ranks it by the highest private score among them. The organisers determine the final standings separately, on independent holdout data recorded with different stimuli; all ranks in this paper are Kaggle's. We relied heavily on

2026 PNPL Competition.

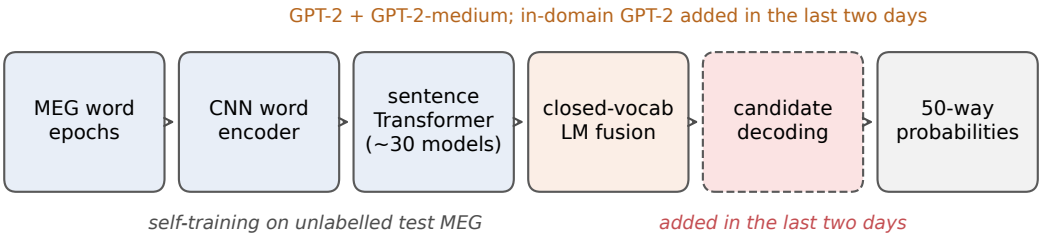


Figure 1: System overview. Word epochs pass through a per-word convolutional network (CNN) and a sentence Transformer; the ensemble's 50-way log-probabilities are combined with a language-model prior and, in the last two days, with whole-sentence candidate decoding (dashed).

the public score, submitting up to ten times a day, which may have exposed us to adaptive over-selection. Kaggle released the private scores of all our submissions, so we can study retrospectively how public-score gains and our final selection related to private performance. In our history, late public-score gains did not carry over to the private split, and our last-day reselection lowered our best private score.

The paper has three parts: the system (Section 2), a retrospective public–private analysis (Section 3), and a back-test of final-selection rules with recommendations (Section 4). All leaderboard statistics are computed from the Kaggle export of our submissions by a released script. Hand-coded inputs, namely which submissions are compared, their labels and the two selections, are listed in that script, and each validation label cites its dated entry in our experiment log.

2 System

Data and validation. A run is one continuous recording, and a session is one recording visit. Each run is z-scored per channel and projected by a global, whitened principal component analysis (PCA) from 306 channels to 80 components. A word epoch is the window around a word's onset: $[-0.5, 2.5]$ s at 50 or 100 Hz ("3 s" models) or $[0, 1]$ s at 125 Hz ("1 s" models). TIMIT SA1/SA2 sentences recur 630 times each, and sentence models memorise them, so we train repeated sentences as isolated words.

Validation uses seven runs held out from supervised training (the PCA was fitted on a sample of runs that included two of them). We report two validation measures on these runs, both in BAcc@10. *Novel-sentence validation* uses held-out sentences whose text never occurs in training. The *proxy* mimics the test sentences with the 312 stretches of three or more consecutive target words in these sentences (1,042 words), so that LM fusion can be tested. The test recordings were made in separate sessions, and we normalise them over their own short segments, so the standard deviation of each PCA component in test epochs is about 1.35 times that in training epochs.

MEG decoder. Figure 1 summarises the pipeline. We follow the word-decoding setup of d'Ascoli et al. [1], without their semantic objective. A convolutional encoder maps each word epoch to an embedding, and a pre-norm Transformer over the words of a sentence classifies each word among 1,000 classes: the 50 target words and the 950 most frequent other words. The submitted probabilities are a softmax over the 50 target classes only. The base configuration has model width $d{=}512$ and 8 Transformer layers, and a second head classifies each word without context. Final ensembles average the log-probabilities of about 30 members, which differ in window, sampling rate, depth (4 to 16 layers), CNN width and input normalisation. IN1 is our 31-member ensemble with two instance-normalised models and one window-normalised model. Some members are self-trained on the unlabelled test recordings, using an earlier submission's predictions as soft targets.

Language prior. For each sentence position, a causal LM gives the probability of each of the 50 words, averaged over the beam hypotheses for the preceding words. Following pointwise mutual information (PMI), we subtract, for each word, the log of its average LM probability over all word positions of the test sentences. The result is added to the MEG log-probabilities with weight 0.5,

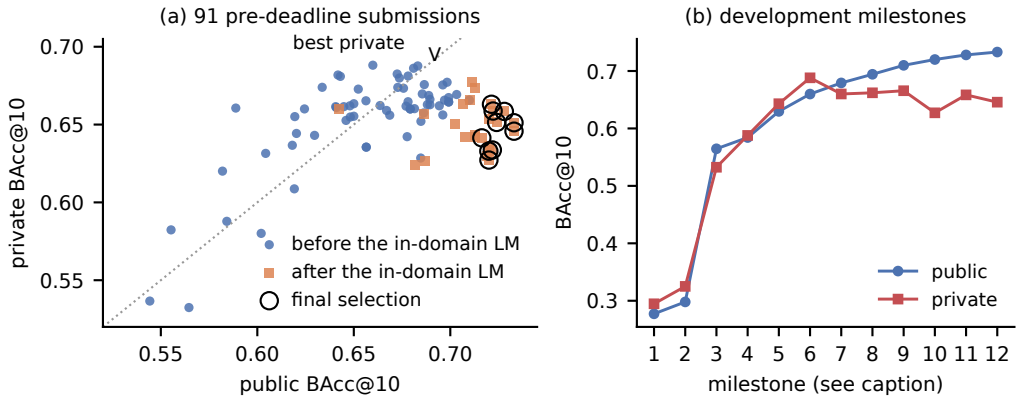


Figure 2: (a) Public and private BAcc@10 of the 91 pre-deadline submissions with public score at least 0.5 (dotted line: equal scores), marked by whether they were made before or after the in-domain LM was introduced on 14 October. Black rings mark our ten final selections. (b) Milestones: 1 linear baseline, 2 per-word CNN, 3 sentence Transformer, 4 repeated-sentence handling, 5 closed-vocabulary LM fusion, 6 the 9-model ensemble with our best private score, 7 3 s windows and LM calibration, 8 self-training, 9 the 28-model ensemble B5, 10 in-domain LM (807 sentences, the lowest of three LM fine-tuning seeds on private), 11 IN1 with the 3,742-sentence LM, 12 candidate decoding. Each milestone may change several components.

using beam search with 32 hypotheses. GPT-2 [3] and GPT-2-medium are combined after rescaling their PMI terms to the same standard deviation.

Prior correction, word timing and V. *Prior correction* with strength τ subtracts τ times the log training frequency of each word from the MEG log-probabilities. *Word-timing evidence* with weight β adds β times the log-likelihood of the gap to the next word's onset given the word. The *validation-motivated stack V* combines input rescaling, word timing ($\beta = 0.5$) and sentence prior correction ($\tau = 0.3$).

Components added in the last two days. The *in-domain LM* is GPT-2 fine-tuned on corpora of 807 to 3,742 synthetic sentences written by Claude and Qwen3.5-9B [4], prompted with the 50 target words and the example utterances of the competition paper [2], and filtered to sentences that use only those words. Two variants use an in-domain Qwen2.5-1.5B instead of or alongside the in-domain GPT-2. *Whole-sentence candidate decoding* scores every candidate sentence with the same number of words as the test sentence by its summed MEG log-probability. A softmax over the candidates gives per-position word probabilities, which are mixed into the submission with weight λ. If fewer than 20 candidates of that length exist, the base prediction is kept. Generic candidates come from the synthetic corpora ($\lambda = 0.15$). Our best public submission adds LLM proposals (guided candidates) prompted with our six highest-scoring words per position, a prior of 0.5 times each candidate's in-domain LM log-probability, and $\lambda = 0.3$.

Isolated words. For the 92 isolated-word test items we averaged the predictions of the contextual and context-free heads, corrected with $\tau = 0.7$, assuming that these rows counted towards the private score only. Four pairs of submissions differed only in these rows (in 33–82% of their top-10 sets, in the three pairs whose files we kept), yet had identical public and private scores to five decimals. Our tuning of these rows therefore had no measurable effect, which suggests that they are not scored.

3 Public versus private

Scores of all submissions. The export contains 99 submissions; we analyse the 95 made before the deadline and exclude throughout four ablations of candidate decoding made afterwards. Over the 95, public and private scores correlate at $r = 0.92$, but this reflects the large early steps. The correlation is 0.50 once the four sub-0.5 submissions are excluded, 0.11 for the 85 submissions with

Table 1: Seventeen of the 30 reference–variant comparisons, covering the components of Section 2 (change in BAcc@10, variant minus reference), with the direction suggested by component-level validation recorded before the private release ($+$ helps, $-$ hurts; n/a: no matched validation difference above the proxy noise of about 0.015). †: bundle of several changes.

Change (variant minus reference)	Δ public	Δ private	Validation
sentence-context model (redesign)†	$+0.267$	$+0.207$	n/a
closed-vocab LM fusion (PMI)	$+0.045$	$+0.055$	$+$
Qwen3-8B instead of GPT-2	-0.041	-0.079	$-$
LM fusion without PMI (N-best)†	-0.074	-0.061	n/a
self-training round 1	$+0.015$	$+0.002$	n/a
LM fusion after self-training	$+0.022$	-0.012	n/a
all members self-trained (one teacher)†	-0.014	-0.049	n/a
sentence-end LM term	-0.051	$+0.019$	n/a
sentence prior correction τ=0.3	-0.037	$+0.017$	$+$
word-timing evidence β=0.5	-0.060	-0.010	$+$
validation-motivated stack V†	-0.027	$+0.022$	$+$
in-domain LM (807 sentences, seed 0)	$+0.010$	-0.038	n/a
in-domain LM, 3-seed mixture†	$+0.010$	-0.012	n/a
in-domain corpus 807 $\rightarrow$ 1,368	-0.010	$+0.038$	n/a
in-domain corpus 807 $\rightarrow$ 2,103 (+ Qwen sentences; IN1)	$+0.002$	$+0.018$	n/a
generic candidates, λ=0.15	-0.003	$+0.012$	n/a
candidate decoding (guided pool + LM prior)†	$+0.009$	-0.006	n/a

public score at least 0.6, and -0.37 for the 60 at or above 0.66 (95% interval -0.57 to -0.13). We chose these thresholds after seeing the data, and this and later intervals are nominal, since they treat submissions as independent. The negative value is a contrast between two groups (Figure 2a). Among the 60, the 24 submitted after the in-domain LM was introduced (from 14 October 16:00 UTC, the last two days) averaged 0.714 public and 0.648 private (standard deviations 0.013 and 0.015), against 0.686 and 0.666 (0.012 and 0.012) for the 36 earlier ones, and within each group the correlation is small (0.17 and 0.01). Our best private score, 0.6881 (public 0.65986), came from a 9-model ensemble submitted on 9 October (Figure 2b).

Why differences between related submissions were unreliable. Three factors may explain why. First, most variants were built on a public leader: in 29 of our 30 comparisons below, the reference had set or tied the public maximum when it was submitted. If its public score partly reflected favourable noise, its variants' public scores should fall back while their private scores should not; the 26 later comparisons are consistent with this (mean change -0.0095 public against -0.0003 private, standard deviations 0.021 and 0.019). Second, self-training and guided candidates copy a public-selected submission's test predictions into later submissions, so the later submissions are not independent. Third, training noise alone moves a score by more than most of our effects. Three repeats of one recipe that re-seeded only the fine-tuning of the 807-sentence in-domain LM, with the MEG ensemble unchanged, gave private scores of 0.627, 0.642 and 0.630 (standard deviation 0.008; nominal 95% interval 0.004 to 0.050). Twice the standard deviation of a difference of two such draws is about 0.022. We use it as a rough scale (the seed scale) rather than a calibrated threshold; 23 of our 30 private effects are below it. For comparisons that change only post-processing of the same trained models, the relevant noise is instead the sampling of test items, which we cannot estimate. Public scores repeat five-decimal values more often than private ones (81 against 90 distinct values among 95 submissions), consistent with a small public split; we know neither its size nor how Kaggle assigned test rows to the splits.

Paired comparisons. We specified 30 reference–variant comparisons between submissions: 15 change one documented component and 15 bundle several (Table 1, Figure 3a, Appendix A). The four early structural changes, made between 7 and 9 October, kept their sign on the private split: the sentence-context model ($+0.207$ private), LM fusion ($+0.055$), and the two LM alternatives we rejected. Among the 26 later comparisons, only 10 kept their sign (Wilson 95% interval 0.22–0.57,

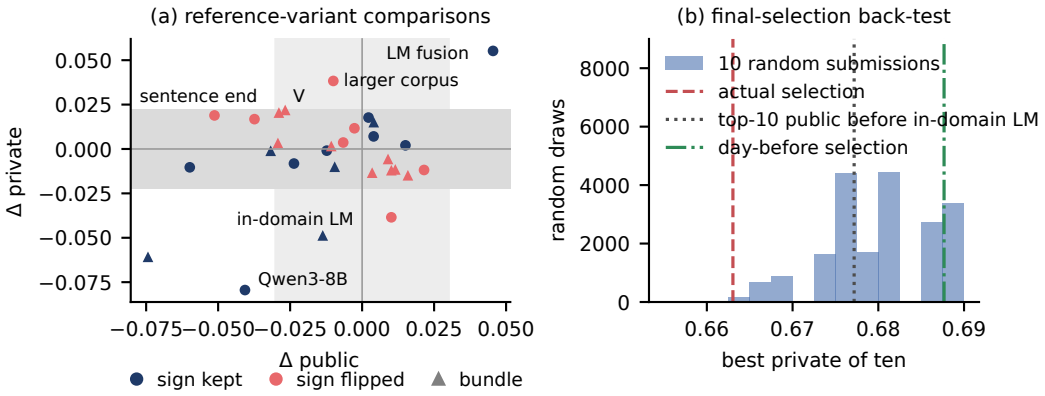


Figure 3: (a) Public and private effect (BAcc@10) of the 30 reference–variant comparisons; the sentence-context redesign ($+0.267/+0.207$) is off the scale. Light grey band: absolute public effect below 0.03; darker band: absolute private effect below 0.022, the seed scale of Section 3, which contains 14 of the 16 sign changes. Light red: the sign changed between splits; triangles: bundles. (b) Best private score of ten distinct submissions drawn at random (20,000 draws) from the 60 pre-deadline submissions with public score at least 0.66, with three selections from Table 2. The distribution is discrete: the best of ten is one of the pool's scores.

which includes one half), and the public and private effects were not positively related ($r = -0.29$, interval -0.61 to 0.11). These counts are consistent with noise of the size above: 23 of the 26 private effects are below 0.022, none of the 26 comparisons exceeds it on both splits, and the four later changes with an absolute public effect of at least 0.03 split two and two, with all four private effects below 0.022. The evidence that the public split misled lies instead in the group contrast above and in the final selection (Section 4). The comparisons share references and were chosen by us.

The language components. Replacing GPT-2 with the 807-sentence in-domain LM increased the public score by 0.010 and decreased the private score by 0.038. All three fine-tuning seeds lost on the private split (-0.038, -0.024 and -0.036; mean -0.033, standard error 0.005), so on this split the loss exceeds the seed scale; a mixture of the three seeds lost 0.012. Larger corpora mostly recovered this loss but did not clearly exceed the reference without the in-domain LM. Adding the 807 Claude sentences to the 1,575 Qwen sentences (2,103 distinct sentences in all) even reduced the private score from 0.674 to 0.651 while raising the public score. The five single-seed 807-sentence variants with the highest public scores (0.719–0.722) scored 0.627–0.636 on private, among the lowest. Generic candidates changed the private score by $+0.012$, and the guided recipe with the best public score by -0.006, both within the seed scale. In short, the 807-sentence LM lowered the private score, the data neither establish nor rule out a benefit of the larger corpora or of candidate decoding, and selecting variants by public score picked poor ones.

Validation against public. For six comparisons, in-session validation and the public score pointed in opposite directions before the private release. Private-score changes agreed with validation twice ($\tau = 0.3$ and V) and with the public score four times ($\tau = 0.5$, the wide CNN and two word-timing strengths). Following validation would have changed the private score by $+0.003$ on average (nominal 95% interval -0.011 to $+0.017$; exact signed-rank $p = 1.0$, assuming independent comparisons), with every individual effect at or below the seed scale (the largest, V's $+0.022$, is at it). The cases also overlap, since V contains $\tau = 0.3$ and word timing. Validation was therefore neither clearly better nor clearly worse than the public score, which we followed during the competition.

4 Final selection

One day before the deadline, our recorded selection mixed four in-domain-LM submissions with three earlier ensembles and three validation-backed submissions, including V. On the last day, after

Table 2: Back-test of final-selection rules: best private score among ten pre-deadline submissions. Random rules: exact expected best, and exact probability that the best is no higher than our actual selection (0.66304).

Rule	Best private	P(best $\leq$ actual)
Our actual final selection	0.663	–
Ten highest public scores (8 of 10 in the actual selection)	0.663	–
Ten highest public before the in-domain LM	0.677	–
Ten random of the 60 with public ≥ 0.66 (expected)	0.680	0.24%
Ten random of those 60 after the in-domain LM (expected)	0.672	9.4%
Our selection one day before the deadline (included V)	0.688	–

the 3,742-sentence LM and candidate decoding raised the public score further, the final selection replaced all six submissions made before the in-domain LM with later ones, which moved the result from 0.6877 to 0.6630; on Kaggle's private leaderboard, V's score would have placed us 34th instead of 44th.

Table 2 and Figure 3b compare simple alternatives. Only 0.24% of random sets of ten from the 60 submissions with public score at least 0.66 were as low; at five tested cutoffs from 0.64 to 0.68 the figure is between 0.15% and 0.33%. Within the 24 submissions made after the in-domain LM, the family that the final selection came from, only four scored above our actual selection (0.66304), so 9.4% of random sets of ten were as low: most of the 0.24% reflects the choice of family rather than the choice within it. V is the pool's best; the overall best (public 0.65986) is just below its cutoff. The pool and rules are defined in hindsight, so these figures describe our submissions rather than a general result.

Our recommendations for small public splits:

1. Repeat at least one recipe with different seeds before the final selection, and treat public or validation differences below the resulting seed scale (about 0.02 for our LM recipe) as weak evidence.
2. Diversify the final selection across imperfectly correlated criteria (public score, validation, earlier strong systems). With best-of-ten scoring this protects against any single criterion being wrong, as our day-before selection would have.
3. Record each submission's reference, and compare variants with a fixed reference rather than the running public maximum, which adds fresh favourable noise to each comparison.
4. Check which test rows are scored, for example with pairs of submissions that differ only in some rows.

5 Discussion

For this single, deeply sampled subject, two large structural changes held up on the private split: the sentence-context redesign ($+0.207$) and closed-vocabulary LM fusion ($+0.055$). Most later private effects were below the seed scale. One further explanation remains open: we do not know how Kaggle assigned test rows to the two splits. A validation proxy built from recordings outside the training sessions would be a promising improvement.

Reproducibility. The repository `https://github.com/kuotunyu/pnpl2026-meg-word-decoding` contains the code, the corpora, the submission export with leaderboard snapshots, and the analysis script, and links to the checkpoints. MEG models were trained on Google Colab A100 GPUs and a local RTX 4090.

Acknowledgments and Disclosure of Funding

We thank the PNPL organisers for the competition and the dataset. The release of private scores for all submissions made this analysis possible. This work received no external funding; the author declares no competing interests.

References

[1] S. d'Ascoli et al. Towards decoding individual words from non-invasive brain recordings. *Nature Communications*, 16, 2025. doi:10.1038/s41467-025-65499-0.

[2] F. Mantegna, G. Elvers, D. Jayalath, G. Landau, T. Kim, M. Özdogan, L. Kurth, T. Kwon, S. Cho, B. Ballyk, A. Fung, A. Greer, P. Somaiya, C. Herff, Y. M. Ramos, H. Abdelhedi, K. Jerbi, G. Farquhar, B. Shillingford, M. Woolrich, O. P. Jones. The 2026 PNPL Competition: Word Classification and Efficient Cross-Subject Generalisation in LibriBrain100. arXiv:2609.03231, 2026.

[3] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever. Language models are unsupervised multitask learners. OpenAI technical report, 2019.

[4] Qwen Team. Qwen3.5-9B. Model card, `https://huggingface.co/Qwen/Qwen3.5-9B`, 2026.

A All reference–variant comparisons

Table 3 lists all 30 comparisons with the Kaggle file names of the reference and the variant. Validation labels are component-level: they come from smaller validation ensembles, were recorded before the private release, and were sometimes recorded after the public result; the experiment-log entry behind each validation label is listed in the released analysis. The higher 1 s share (comparison 8) is labelled n/a because its proxy evidence also added four models, while novel-sentence validation changed by only 0.004. Some references were submitted after their variant, as matched controls. With an absolute public-effect threshold of 0.025, 0.03 or 0.035, the large and small groups kept their sign in 6 of 11 and 8 of 19, 6 of 8 and 8 of 22, or 5 of 7 and 9 of 23 cases. The ranks come from leaderboard snapshots saved on 29 October 2026; the public leaderboard listed 90 ranked teams and the hosts' unranked baseline. Our two recorded selections are, on 15 October: `IN1_Q_llm_plus`, `B5i_L1_IN1`, `B5i_L1`, `B9c_L1`, `B5i`, `B9c`, `M1`, `B5d10`, `V`, `F1`; and on 16 October (final): `AVG_SCGL_Qbig`, `IN1Qp_SCGL_b0.5`, `IN1_Q_big`, `IN1_Q_llm_plus`, `B9cIN_Qplus`, `B5i_L1_IN1`, `IN1Qp_SC0.15`, `B5i_L1`, `B9c_L1`, `IN1_Q_big_SCG2L_l0.5` (file names without the `final_` prefix). The tables and statistics are generated by `analysis/analyze_lb.py` and `analysis/make_tables.py` from `analysis/data/submissions_raw.csv`.

Table 3: All 30 reference–variant comparisons (change in BAcc@10, variant minus reference). Type: single documented change or bundle of several. Validation as in Table 1.

#	Change	Reference	Variant	Type	Δ public	Δ private	Validation
1	sentence-context model (redesign)	`sub_v1`	`sub_c1`	bundle	$+0.267$	$+0.207$	n/a
2	closed-vocab LM fusion (PMI)	`sub_c5`	`sub_c5_lmpmi05`	single	$+0.045$	$+0.055$	$+$
3	LM fusion without PMI (N-best)	`sub_c5_lmpmi05`	`sub_c5_lm05`	bundle	-0.074	-0.061	n/a
4	Qwen3-8B instead of GPT-2	`final_gpt2`	`final_qwen3_8b`	single	-0.041	-0.079	$-$
5	3 s window ensemble	`final_gpt2`	`finalW_gpt2`	bundle	$+0.016$	-0.015	n/a
6	GPT-2-medium added, strength-calibrated	`finalW_gpt2`	`finalW_calib2`	bundle	$+0.003$	-0.013	n/a
7	sentence prior correction τ=0.5	`finalW_calib2`	`finalW_calib2_t0.5`	single	-0.012	-0.001	$+$
8	higher 1 s share	`finalW_calib2_iso14`	`finalY_calib2_t0`	bundle	-0.029	$+0.003$	n/a
9	self-training round 1	`finalW_calib2_iso14`	`final_P1`	single	$+0.015$	$+0.002$	n/a
10	LM fusion after self-training	`final_P1noLM`	`final_P1`	single	$+0.022$	-0.012	n/a
11	self-training round 2 (all replaced)	`final_P1`	`final_P2`	bundle	-0.010	-0.010	n/a
12	sentence-end LM term	`final_P1`	`final_P1eos`	single	-0.051	$+0.019$	n/a
13	larger diverse ensemble	`final_P1`	`final_B2`	bundle	$+0.004$	$+0.015$	n/a
14	12-layer + wide-CNN members	`final_B2`	`final_B5`	bundle	$+0.012$	-0.012	n/a
15	all members self-trained (one teacher)	`final_B2`	`final_C`	bundle	-0.014	-0.049	n/a
16	more 1 s members	`final_B5`	`final_B6a`	bundle	-0.029	$+0.021$	n/a
17	wide CNN instead of 12-layer	`final_B5`	`final_F1`	bundle	-0.032	-0.001	$+$
18	sentence prior correction τ=0.3	`final_B5i`	`final_B5i_t0.3`	single	-0.037	$+0.017$	$+$
19	word-timing evidence β=0.5	`final_B5i`	`final_B5d05`	single	-0.060	-0.010	$+$
20	word-timing evidence β=1.0	`final_B5i`	`final_B5d10`	single	-0.024	-0.008	$+$
21	input rescaling	`final_B5i`	`final_B5irs`	single	-0.007	$+0.004$	n/a
22	validation-motivated stack V	`final_B5i`	`final_V`	bundle	-0.027	$+0.022$	$+$
23	12-layer members, lower learning rate	`final_B5i`	`final_B8b`	bundle	-0.011	$+0.002$	n/a
24	in-domain LM (807 sentences, seed 0)	`final_B5i`	`final_B5i_L1`	single	$+0.010$	-0.038	n/a
25	in-domain LM, 3-seed mixture	`final_B5i`	`final_B5i_L1avg`	bundle	$+0.010$	-0.012	n/a
26	in-domain corpus 807 $\rightarrow$ 1,368	`final_B5i_L1`	`final_B5i_L3`	single	-0.010	$+0.038$	n/a
27	in-domain corpus 807 $\rightarrow$ 2,103 (+ Qwen sentences; IN1)	`final_B5i_L1_IN1`	`final_IN1_Q_llm_plus`	single	$+0.002$	$+0.018$	n/a
28	in-domain corpus 2,103 $\rightarrow$ 3,742 (IN1)	`final_IN1_Q_llm_plus`	`final_IN1_Q_big`	single	$+0.004$	$+0.007$	n/a
29	generic candidates, λ=0.15	`final_IN1_Q_llm_plus`	`final_IN1Qp_SC0.15`	single	-0.003	$+0.012$	n/a
30	candidate decoding (guided pool + LM prior)	`final_IN1_Q_llm_plus`	`final_IN1Qp_SCGL_b0.5`	bundle	$+0.009$	-0.006	n/a